

Tilen Žel

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📍 Location: Ljubljana, Slovenia

🐙 Github: github.com/TZ387

EDUCATION

University of Ljubljana

Faculty of Mathematics and Physics

Master's Degree (Physics)

SKILLS

Languages and platforms:

C/C++  , Python  , Matlab  , Julia  , CUDA 

Other:

Microsoft Office, LaTeX/Typst, Medical Physics, Photonics

WORK EXPERIENCE

Fotona

R&D Engineer

Ljubljana

Sep 2021 - Present

- Made major improvements in area of numerical simulations, which has led to significant benefits for research and regulatory approval of various devices.
- Participated in research projects involving laser technology, laser treatments, and laser-tissue interaction

Faculty of Mechanical Engineering (FS)

Research Aide

Ljubljana

Oct 2018 - July 2021

- Assisted in developing a simulation of capillary tubes, which are a crucial component of various cooling devices, such as air conditioners and refrigerators.
- Analyzed the condensation of moist air in water condensers for different geometries and flow regimes.

Jozef Stefan Institute (IJS)

Research assistant (Master's degree)

Ljubljana

Dec 2019 - June 2021

- Analyzed spectroscopic measurements and optically scattering materials by using and modifying the Monte Carlo method in the CUDA platform and the C programming language.

PROJECTS

Magnetic simulation for arbitrary coils (Fotona): Matlab, C, CUDA

Modified a Matlab toolbox for calculation of magnetic field, improved it to enable calculation of induced current in tissue, and accelerated it by enabling its computation on graphic card using C and CUDA.

Simulation of capillary tubes (FS): Matlab, C++, wxWidgets

My work consisted of minor assistance in the creation of the original code in Matlab and re-implementation of the entire project to the C++ programming language, along with creation of a GUI.

Laser resonator simulation using 2D FFT (Fotona): Julia, CUDA

Developed a simulation of laser cavity dynamics for both stable and unstable resonators using Fourier optics beam propagation, with optional GPU acceleration.

Further projects (Python, JavaScript, Matlab, Julia, QML, etc.) are available on my GitHub page.

PUBLICATIONS

J M. Lukač, J. Košir, T. Žel, M. Kažič, D. Šavli and M. Jezeršek, "Influence of tissue desiccation on critical temperature for thermal damage during Er:YAG laser skin treatments", *Lasers in Surgery and Medicine*, No. 56(1), pp. 107-118, Nov. 2023, doi: <https://doi.org/10.1002/lsm.23739>

C B. Majaron and T. Žel, "Optical properties of Spectralon assessed by replication of literature data in Monte Carlo simulations", *Proceedings Volume 12147, Tissue Optics and Photonics II; 1214704*, Strasbourg, France, May 2022, doi: <https://doi.org/10.1117/12.2620704>

J P. Poredoš *et al.*, "Condensation of water vapor from humid air inside vertical channels formed by flat plates", *iScience* 25(1), pp. 1--26, Jan. 2022, doi: <https://doi.org/10.1016/j.isci.2021.103565>